Business Requirements Document (BRD)

Project Name: AI-Powered E-Commerce Platform (FastAPI + LangGraph)

Document Version: 1.0

1. Project Ka Maqsad (Project Overview)

Is project ka maqsad ek basic open-source e-commerce repository ko clone kar ke usay ek fully functional, production-ready platform mein tabdeel karna hai. Is system mein custom payment methods (Bank Account + COD), ek naya Admin Dashboard, aur sab se eham: **5 advanced LangGraph-based AI Agents** shamil honge jo backend (FastAPI) par mukhtalif tasks ko automate karenge. Akhir mein isay Render par deploy kar ke professional portfolio ka hissa banaya jayega.

2. Step-by-Step Execution Plan

Phase 1: Core E-Commerce Setup & Customization

1. **Repository Cloning & Setup:** GitHub se ek suitable e-commerce project clone kar ke locally run karna then Django sy logic fastapi mai convert krna hy. https://github.com/msncoder/Django-Trendora
2. **Payment System Modification:** Maujooda online payment gateway (e.g., Stripe/PayPal) ko hata kar manual "Bank Account Number" transfer aur "Cash on Delivery (COD)" ka option lagana.
3. **UI & API Enhancements:**
    - Frontend (UI) ko user-friendly aur responsive banana.
    - FastAPI ke endpoints ko optimize karna taake response time behtar ho.
4. **Admin Dashboard Development:** Agar cloned repo mein admin panel nahi hai, to ek naya dashboard banana jahan se products, orders, aur users ko manage kiya ja sake.

Phase 2: Quality Assurance (UAT)

- **User Acceptance Testing (UAT):** AI integrate karne se pehle poore system ki testing karna. Is mein cart functionality, checkout process, admin controls, aur payment flows ko check kiya jayega taake base system mein koi bug na rahe.

Phase 3: AI Agents Integration (FastAPI + LangGraph)

UAT pass hone ke baad, neechay diye gaye 5 AI agents ko system mein integrate kiya jayega:

Agent 1: Smart Refund & Escalation Agent (Conditional + Human-in-the-loop)

- **Yeh kia karega:** Jab koi customer refund ya order cancel karne ki request karega, to yeh agent NLP ke zariye uski baat ka matlab samjhega aur SQLAlchemy ke through uski order

history nikalega.

- **Kaam karne ka tariqa (Workflow):**
  - **Sequential:** Pehle intent analyze karega, phir order details fetch karega.
  - **Conditional:** Agar amount thori hai aur policy ke mutabiq hai, to khud hi refund process kar dega.
  - **Human-in-the-loop:** Agar amount bari hai ya case complex hai, to graph pause ho jayega (interrupt_before). Phir admin dashboard par notification aayega, jahan se admin FastAPI ke /approve-refund endpoint par click kar ke request ko approve ya reject karega.

Agent 2: Dynamic Product Discovery & Recommendation Engine (Parallel + Reducers)

- **Yeh kia karega:** Customers ki mushkil search queries (jaise "sardi ke liye jacket aur shoes under 5000") ko samajh kar behtareen products nikal kar dega.
- **Kaam karne ka tariqa (Workflow):**
  - **Parallel Execution:** Agent search ko hisson mein baant dega. Ek saath teen (3) parallel branch chalengi: ek jackets dhoondegi, doosri shoes dhoondegi, aur teesri user ki purani purchase history check karegi.
  - **State Reducers:** In teeno jaghon se aane walay data ko ek custom reducer function merge karega. Filter aur sort karne ke baad final list banayega jo FastAPI ke /recommendations endpoint ke zariye UI par show hogi.

Agent 3: Competitor Pricing & Inventory Auditor Agent (Iterative / Tool-Routing)

- **Yeh kia karega:** Yeh background mein kaam karega aur competitors ki websites par ja kar aapke products ki prices check karega. Phir aapko mashwara dega ke price kam karni chahiye ya barhani chahiye.
- **Kaam karne ka tariqa (Workflow):**
  - **Iterative Loop:** Agent ko ek product assign hoga. Woh Tavily (ya koi aur search tool) use kar ke web par price dhoondega.
  - Ek reviewer_node check karega ke "kya competitor ki sahi price mil gayi hai?". Agar data ghalat hua, to agent wapas loop mein ja kar dobara search karega. Jab sahi data mil jayega tabhi loop break hoga aur admin ko price update ka suggestion milega.

Agent 4: Order Fraud Detection & Verification Pipeline (Sequential)

- **Yeh kia karega:** Har naye order par nazar rakhega taake fake orders ya fraud se bacha ja sake.
- **Kaam karne ka tariqa (Workflow):**
  - Jaise hi customer /checkout karega, background mein yeh sequential graph chal padega.
  - Pehla node IP address check karega -> Doosra node payment method verify karega -> Teesra node dekhega ke user ne kitni jaldi order place kiya hai.
  - Har node apna ek "fraud score" add karega. Aakhir mein total score ki base par order ko "Safe" ya "Flagged" mark kar diya jayega (Flagged orders admin dashboard mein review ke liye jayenge).

Agent 5: Automated Product Onboarding & SEO Agent (Iterative)

- **Yeh kia karega:** Vendor ka kaam asaan karega. Vendor sirf product ki picture aur naam dega, baki sab yeh agent khud likhega (Description, SEO tags, category).
- **Kaam karne ka tariqa (Workflow):**
  - Image upload hone par pehla node Gemini vision model use kar ke image se details (rang, material, style) extract karega.
  - Doosra node us detail se ek SEO-friendly description banayega.
  - Ek validation_node check karega ke kya description platform ki rules (length, keywords) ke mutabiq hai? Agar nahi, to yeh wapas generation_node ko bheje ga taake aur behtar likha jaye (Iterative refinement).

Phase 4: Deployment & Professional Handover

1. **Cloud Deployment:** Frontend, FastAPI backend, aur PostgreSQL database ko **Render** (ya kisi munasib cloud platform) par live deploy karna.
2. **Assignment Submission:** Project repository ka link, live URL, aur yeh BRD document assignment ke tor par submit karna.
3. **LinkedIn Branding:**
   - Ek detailed LinkedIn post likhna jismein is project ki technical complexities (LangGraph, State Management, Human-in-the-loop) aur real-world use cases ko highlight kiya gaya ho.
   - LinkedIn profile ke 'Projects' section mein is website aur agents ki detail, tech stack, aur live link add karna.

Tech Stack Required:

- **Backend:** FastAPI, Python, SQLAlchemy, PostgreSQL.
- **AI/LLM:** LangGraph, LangChain, Google Gemini (via API), Tavily (for web search).
- **Frontend:** (Depends on the cloned repo - React/Vue/HTML-JS).
- **Deployment:** Render, Docker (optional).